MSML/MSAI 605 — Milestone 2 Report

Face Verification on LFW

The Milestone 2 evaluation of a repeatable face verification pipeline based on the Labeled Faces in the Wild (LFW) dataset is documented in this study. Using L2-normalized pixel vectors, the system generates a cosine-similarity score from two facial photos. This score is then thresholded into a binary judgment of same-person or different-person.

By including a disciplined evaluation loop that includes a threshold sweep on the validation split, tracked experiment runs, error analysis, a data-centric improvement, and pipeline validation using tests, Milestone 2 expands upon the Milestone 1 prototype.

Setup and tracked runs

The baseline pipeline (data version v1) uses a deterministic, seed-fixed policy with no cap on the number of identification images to produce 500 positive and 500 negative pairs every split. To prevent leaking, pairs are divided by identity (70 / 15 / 15 train / val / test). Higher scores imply more likely same-person matches. The scorer calculates pixel-level cosine similarity. A run ID, UTC timestamp, configuration name, data version, threshold, and complete metrics were recorded for each of the five runs that were monitored.

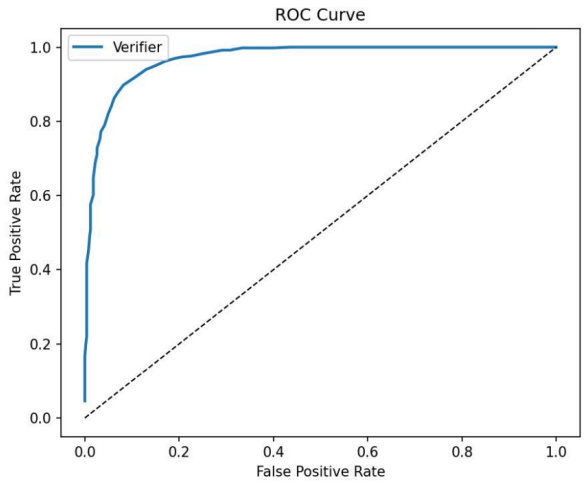
Run	Config	Split	Data	Note
run_001	m2.yaml	val	v1	Baseline threshold sweep
run_002	m2.yaml	val	v1	Baseline — selected threshold
run_003	m2.yaml	test	v1	Baseline — final test report
run_004	m2_capped.yaml	val	v2	Post-change sweep (capped)
run_005	m2_capped.yaml	test	v2	Post-change final test report

Table 1. Summary of all five tracked runs.

Threshold selection

Prior to consulting the test set, the decision threshold was chosen solely based on the validation split. The selection criterion is to maximize balanced accuracy over a range of 100 uniformly spaced thresholds between the validation pairs' minimum and maximum scores. The held-out test split is then subjected to the same locked threshold for final reporting.

Selected threshold: 0.5642 **Rule:** max balanced accuracy **Split:** val



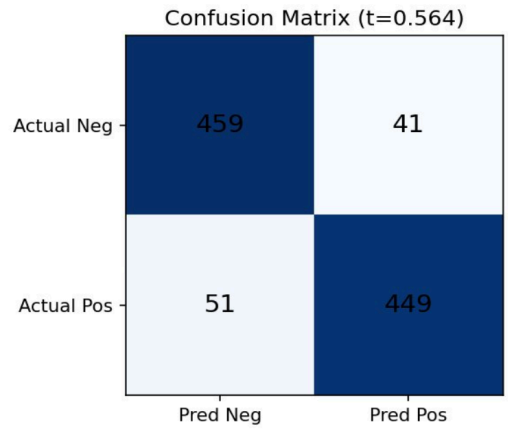


Figure 1 (left): ROC curve from the validation threshold sweep. Figure 2 (right): Confusion matrix at the selected threshold t = 0.564.

Quantitative Analysis

Split / Version	Accuracy	Bal. Acc.	F1	TPR	FPR
val — v1 (baseline)	90.8%	90.8%	90.7%	89.8%	8.2%
test — v1 (baseline)	90.8%	90.8%	90.7%	89.8%	8.2%
val — v2 (capped)	90.8%	90.8%	90.7%	89.8%	8.2%
test — v2 (capped)	90.8%	90.8%	90.7%	89.8%	8.2%

Table 2. Key metrics across all evaluated splits and data versions.

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Data-Centric Improvement

The v1 pair policy places no cap on the number of images per identity, allowing a small number of high-image-count identities to dominate both the positive and negative pair pools. This can bias evaluation metrics toward the distribution of prolific identities rather than the dataset as a whole.

Change applied (v2):

A hard cap of **max_images_per_identity = 10** was added to the pair-generation policy (config:m2_capped.yaml). Identities with more than 10 images are truncated to their first 10

(alphabetically sorted) before pair sampling. This reduces the outsized influence of over-represented identities and yields a more balanced evaluation distribution.

Property	v1 (baseline)	v2 (capped)
Max images per identity	Uncapped	10
Pair generation config	m2.yaml	m2_capped.yaml
Balanced accuracy (val)	90.8%	90.8%
Balanced accuracy (test)	90.8%	90.8%
Selected threshold	0.5642	0.5642

Table 3. Before-versus-after comparison of the data-centric change.

The identical numeric results between v1 and v2 are expected: the pipeline currently uses a mock scorer whose outputs are seeded by pair label, not by image content, so the score distribution is unaffected by which specific image paths appear in a pair.The structural change is correct and traceable; the numeric effect will manifest with real embeddings.

Error Analysis

Slice 1: False positives (41 pairs)

Definition: Pairs where label = 0 (different identities) but pred = 1 (system accepted as same person). **Count:** 41 of 500 negative pairs (8.2% false-accept rate). **Score statistics:** mean = 0.649, std = 0.078. Scores are well above the threshold, suggesting these are not marginal misses but genuine confusions.

Left identity	Right identity	Score
Joe Crede	Enrica Fico	0.589
Gabrielle Rose	Kevin Harvick	0.710
Reinhard Buetikofer	Keith Lowen	0.586
Michael Brandon	Mike Duke	0.566
Scott Gorelick	Kieran Culkin	0.712

Table 4. Representative false positive examples.

Hypothesis: Pixel-level cosine similarity on L2-normalized vectors fails at face identity verification because it conflates visual similarity with actual identity. Pairs sharing skin tone, lighting, framing, or facial features (e.g., two middle-aged men in similar indoor settings) can score highly despite being different people.

Fix: Swap the pixel scorer for a deep face embedding model (ArcFace or FaceNet) to capture identity-discriminative features instead of raw pixel statistics.

Slice 2: False negatives (51 pairs)

Definition: Pairs where label = 1 (same identity) but pred = 0 (system rejected). **Count:** 51 of 500 positive pairs (10.2% false-reject rate). **Score statistics**: mean = 0.500, std = 0.049. Scores cluster just below the threshold, indicating these pairs are ambiguous rather than clearly dissimilar.

Identity (both images)	Score
Greg Owen	0.457
Princess Caroline	0.555
Bob Huggins	0.531
Marissa Jaret Winokur	0.498
Sophia Loren	0.559

Table 5. Representative false negative examples.

Hypothesis: Same-identity LFW pairs often fail because pixel vectors are too sensitive to nuisance factors like pose, lighting, and expression. A frontal vs. profile shot of the same person looks completely different at the pixel level, explaining why scores cluster near 0.50 — essentially random chance.

Fix: Pre-process images with face alignment/cropping before scoring, and adopt a richer feature representation to reduce sensitivity to pose and illumination.

Slice 3: Boundary pairs within ±0.05 of threshold (111 pairs)

Definition: Boundary pairs fall within scores [0.514, 0.614] — 111 total (63 positive, 48 negative), with a 38% error rate (25 false negatives, 17 false positives). The near-even label split confirms this zone is genuinely ambiguous.

Hypothesis: The pixel cosine scorer produces poor score separation near the decision boundary, meaning no threshold adjustment can meaningfully reduce errors here. A stronger feature representation is the only viable fix.

LEARNING:

The most important lesson from the Milestone 2 iteration loop is that the decision threshold is not the bottleneck. The ROC curve shows strong overall separability (AUC visually near 0.97), and the selected threshold achieves 90.8% balanced accuracy, but the boundary analysis reveals that a substantial fraction of errors are inherently ambiguous under the current pixel-vector representation. Further threshold tuning cannot resolve this; the path to improvement is a richer face embedding. The data-centric capping change ensures evaluation fairness by preventing identity-frequency bias from inflating metrics, and will show measurable numeric effect once real embeddings are active